

Manufacturing ELECTRONIC WEIGHING SCALES

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*This report explains
in detail how to set
up an electronic
weighing scale
manufacturing
business unit*

An electronic weighing scale is a sophisticated weighing device, installed with a suite of software to record and process the results received while weighing. It is better than a mechanical scale in various ways. An electronic weighing scale can display readings in numerical form, which allows users to read the measurement results quickly.

In electronic weighing scales, generally, a signal from the load cell is amplified using an analogue-to-digital converter (ADC). It provides output in digital format for displaying, printing, processing and so on. Almost every industry, including retail, manufacturing, laboratory, jewellery, food processing, healthcare, courier companies and weighbridges, uses electronic weighing scales.

Market possibilities

In recent years, the market has experienced an increasing trend in weighing scales due to higher accuracy and lower cost of production. Total production in industrial electronic and control instrumentation is showing growth every year. This is because measuring weight using a digital weighing machine is user-friendly and provides a variety of applications. This product requires a good marketing set up. Most importantly, it must be duly backed by after-sales service facilities.

the assembled PCBs (for performance), electronics assembly, electromechanical assembly, and hardware such as connectors/switches and terminals display. Finally, the assembled unit is calibrated and tested as per the design specification.

Keep the following pointers in mind:

1. Quality standard: IS:2281/1981
2. Production capacity: 1800 units per annum
3. Motive power: 5kVA

International Organisation of Legal Metrology (OIML) standards R 60 and R 76 are the most widely followed by the manufacturers of load cells. In addition, certification ensures that the products meet international specifications for metrological performance and testing.

Waste management practices

The government accords utmost importance to control environmental pollution. A notification for detailed rules to regulate ODS phase-out under Environment Protection Act, 1986 has been put in place with effect from July 19, 2000. The following steps are suggested that may help control the pollution in the electronics industry, wherever applicable:

Fumes and gases are released during hand/wave/dip soldering, which is harmful for people as well as the environment and end products. Alternate technologies may be used to phase out existing polluting technologies. Numerous fluxes have been developed containing two to ten per cent solids, as opposed to the traditional 15 to 35 per cent.

The electronics industry uses CFC, carbon tetrachloride and methyl chloroform for cleaning PCBs after assembly to remove flux residues left after soldering, and various kinds of foams for packaging. Many alternative solvents could replace CFC-113 and methyl chloroform in electronics cleaning.

Chlorinated solvents such as trichloroethylene, perchloroethylene and methylene chloride are being used as effective cleaners for many years. Organic solvents such as ketones and alcohols are effective in removing both solder fluxes and many polar contaminants.

Business compliances

In the beginning, a manufacturer must apply for different registrations and licences from relevant government authorities. However, it is advisable to check the state law.

1. Register the business with Registrar of Companies (ROC).
2. Apply for trade licence from local municipal authority.
3. Apply for MSME Udyog Aadhaar for small and medium unit operations. The same can be registered online.
4. Apply for BIS certification to get ISI mark. Always maintain IS: 2281/1981 quality standard.
5. Obtain NOC from Pollution Control Board. First, apply for Consent to Establish and then for Consent to Operate.
6. Both quality and process are important in this business. ISO 9000 defines the standards for quality management systems. ISO 14001 defines the standards for environmental protection systems for acceptability at international level. Therefore adopt these standards for global competition.
7. Apply for GST registration.
8. Protect brand name with trademark registration.

Manufacturing process

The manufacturing process involves assembly of the load cell, electronic circuits and electromechanical hardware. Subsequently, it includes putting together the electronics assembly, including integrated circuits (ICs), transistors, diodes, resistors, capacitors, transformers, coils, relays and potentiometers, on printed circuit boards (PCBs). However, this must be done as per the design.

Then comes testing